The Solutions for EE4.18 and AO12, 2023

Model answer to Q 1(a): Applied Bookwork and Discussions in Class

This question relates to radiometry for passive millimetre-wave imaging.

i) Briefly describe the basic concept behind a passive radiometric imager and state how this differs from a traditional optical camera.

[3]

A passive radiometric imager is essentially a noise power receiver. With a single detector, different levels of noise radiated by a body is measured at near infrared or longer wavelengths. When a scanning mechanism is added then a 2D noise image can be created. This is different from a traditional optical camera that detects individual photons from a 2D array of detectors in the visible part of the electromagnetic spectrum.

ii) State a suitable millimetre-wave frequency for operating a passive radiometer for outdoor security applications and briefly explain your choice of frequency.

[3]

For security applications, radiometric imaging systems generally operate below 600 GHz, where they can pass through clothing without too much scattering off the clothing. However, if the frequency is too low then there is not enough bandwidth for sufficient noise power to be detected. Ideally, the radiometer should operate in a transmission window within the electromagnetic spectrum, where there is less atmospheric noise and signal attenuation. For this reason, 38, 96, 140 and 250 GHz are suitable frequencies.

iii) Briefly explain the relationship between millimetre-wave frequency and cost for both passive components and active devices.

[2]

In general, at millimetre-wave frequencies, the cost of both passive and active components increased dramatically. With the former, the physical accuracy of manufacturing get harder and so more expensive manufacturing techniques need to be adopted. With the latter, more exotic semiconductor materials are needed to maintain minimum performance requirements.

iv) With a body scanning radiometer, state examples of the types of materials that can be detected beneath a person's clothing?

[2]

Capable of detecting metals, plastic, liquids, gels, ceramics and narcotics concealed beneath a person's clothing.

v) Give two examples of satellite-based applications for passive radiometry.

[2]

Radio astronomy for looking at stars and distant galaxies and earth observation for looking at crop growth and deforestation.

Answers

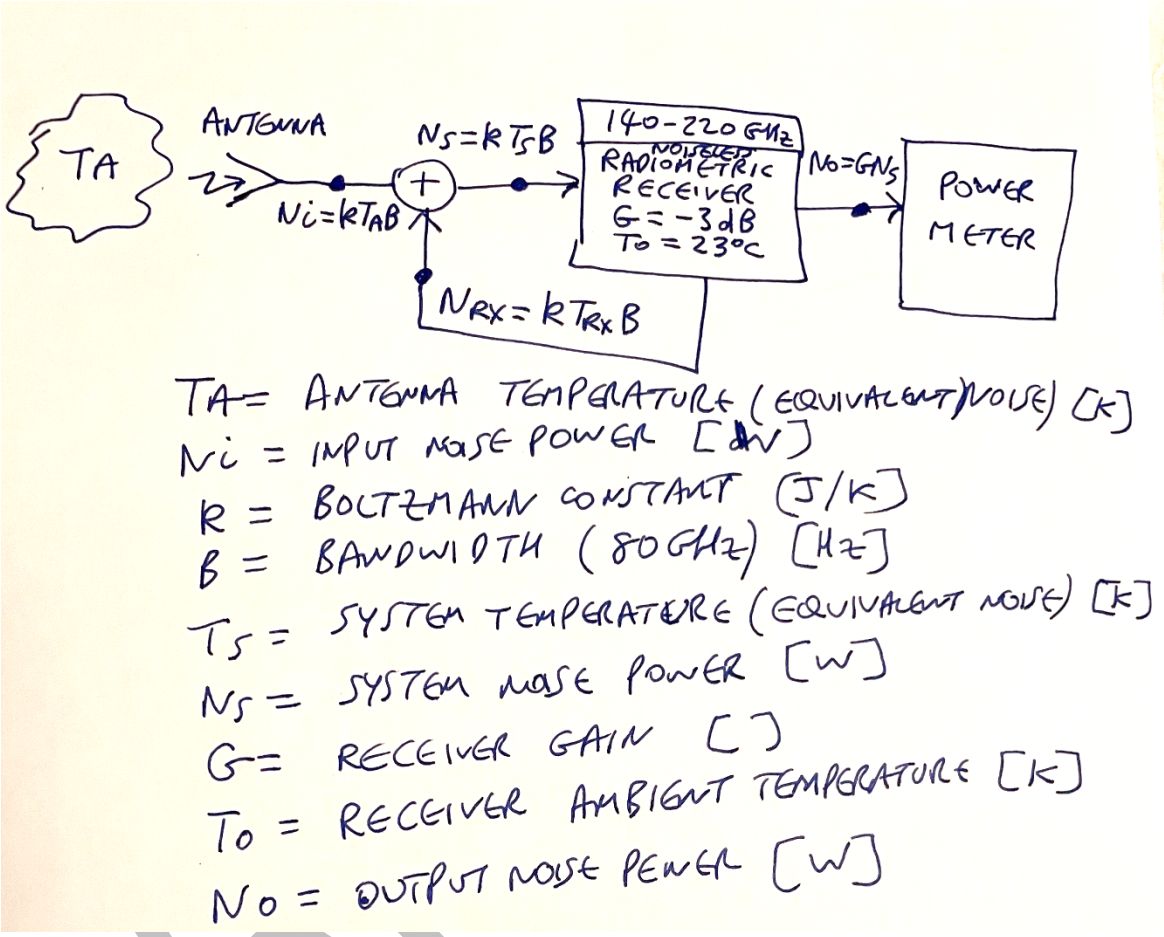
Model answer to Q 1(b): Applied Bookwork

A badly designed radiometric receiver could be just a passive band-pass filter. At its input is an ideal antenna and at its output is a power meter. The relevant specifications for this system are:

Frequency Range: 140 to 220 GHz; Radiometric Receiver Gain: -3 dB;
Radiometric Receiver Ambient Temperature of Operation: 23°C.

Sketch a basic illustration of the complete radiometric system and show all relevant parameters (specifications and variables used in subsequent calculations). Make sure to clearly define all parameters and their units.

[10]



Model answer to Q 1(c): New Calculations

The Boltzmann constant k = 1.380649 ×10^{-23} [J/K] and temperature conversion from Celsius to Absolute T [K] = 273.15 + T [°C]. The radiometric system in 1(b) is used to observe the Sun, having a surface temperature of 5,778 K. With information in 1(b), calculate:

i) Radiometer system bandwidth.

[2]

ii) Radiometric receiver noise figure.

[2]

iii) Input noise power from the antenna, in dBm.

[3]

iv) Radiometric receiver equivalent noise temperature.

[4]

v) Radiometric receiver internal noise power, in dBm.

[2]

vi) Radiometer system equivalent noise temperature.

[2]

vii) Radiometer system noise power, in dBm.

[2]

viii) Output noise power into the power meter, in dBm.

[4]

ix) Required sensitivity for the power meter with 50 Ω input impedance.

[4]

(i) $B = 220 - 140 = 80\,GHz$

(ii) $F_{RX}[dB] \equiv -G[dB] = +3\,dB$

$\therefore F_{RX} = 2$

(iii) $T_A = 5,778\,[K]$

$\therefore N_i = kT_AB \Rightarrow -51.73\,dBm$

(iv) $T_{RX} = T_0(F_{RX} - 1)$

$T_0 = 273.15 + 23 = 296.15[K]$

$\therefore T_{RX} = T_0$

(v) $N_{RX} = kT_{RX}B \Rightarrow -64.85\,dBm$

(vi) $T_S = T_A + T_{RX} = 6,074.15[K]$

(vii) $N_S = kT_SB \Rightarrow -51.95\,dBm$

(viii) $N_O = GN_S \Rightarrow -54.74\,dBm$

(ix) $V_{SC(PEAK)} = \sqrt{2 \times 50\Omega \times N_O} = 819\,\mu V$

Model answer to Q 1(d): Calculations

Assuming that the noise figure in 1(c) does not change, how can the radiometric receiver be simply redesigned so that it can be used with a power meter having a minimum detectable power level of -25 dBm?

[3]

With an output for of only -54.74 dBm, an additional noiseless amplifier can be inserted between the antenna and radiometric receiver, having at least +30 dB of power gain.

Model answer to Q 2(a): Textbook and Calculated Example

(i) For a lossless linear two-port network, state the simple equation that shows the relationship between insertion loss and return loss.

$$Insertion\ loss = -10log_{10}|S_{21}|^2$$

$$Return\ Loss = -10log_{10}|S_{11}|^2$$

$$For\ a\ lossless\ linear\ 2-port\ network, 1 \equiv |S_{21}|^2 + |S_{11}|^2$$

[3]

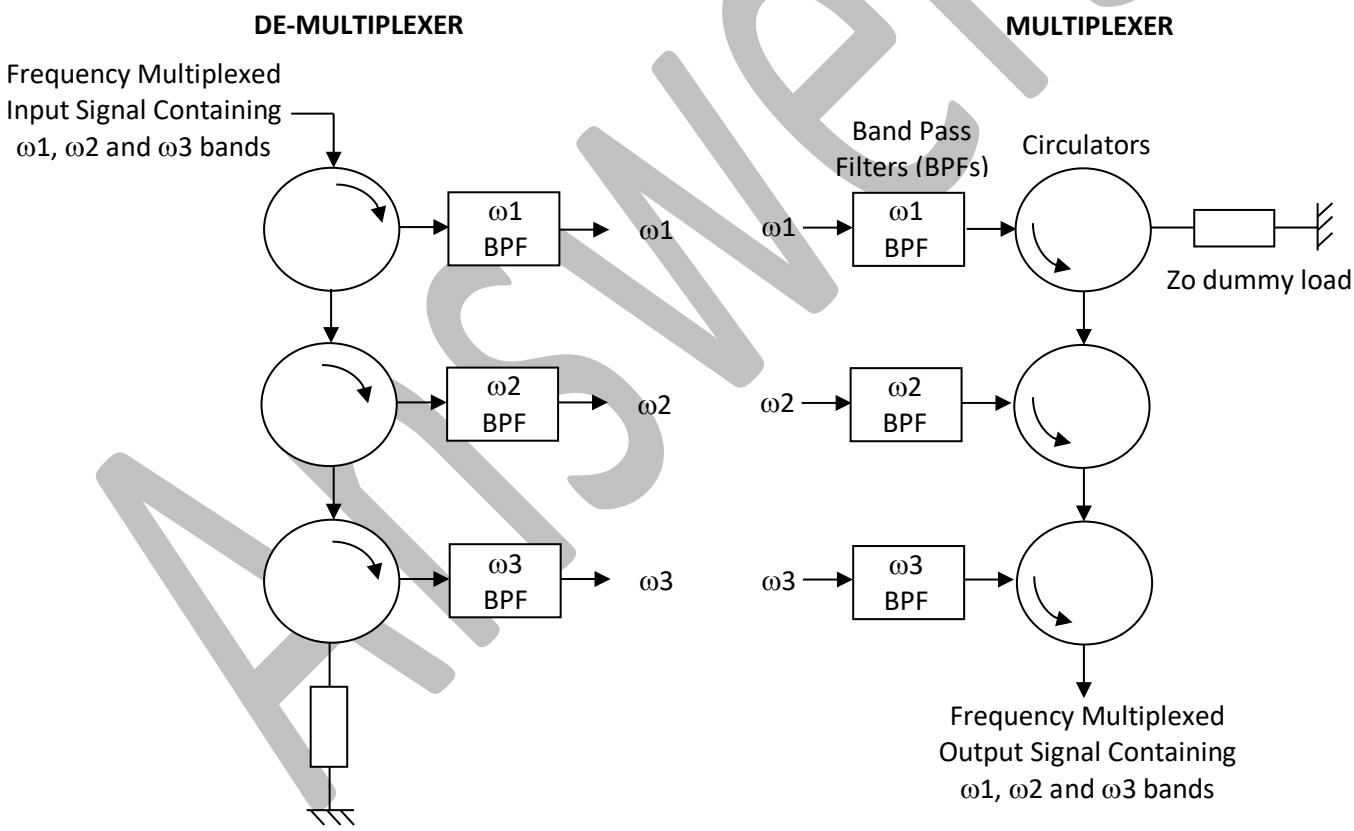
(ii) Using the equation stated in 2(a)(i), calculate the return loss when insertion loss is 1 dB.

$$|S_{21}|^2 = 0.794\ and\ |S_{11}|^2 = 0.206\ and\ Return\ Loss = 6.868\ dB$$

[3]

Model answer to Q 2(b): New Application

Draw a 3-channel frequency de-multiplexer and a 3-channel frequency multiplexer, employing circulators and band-pass filters.

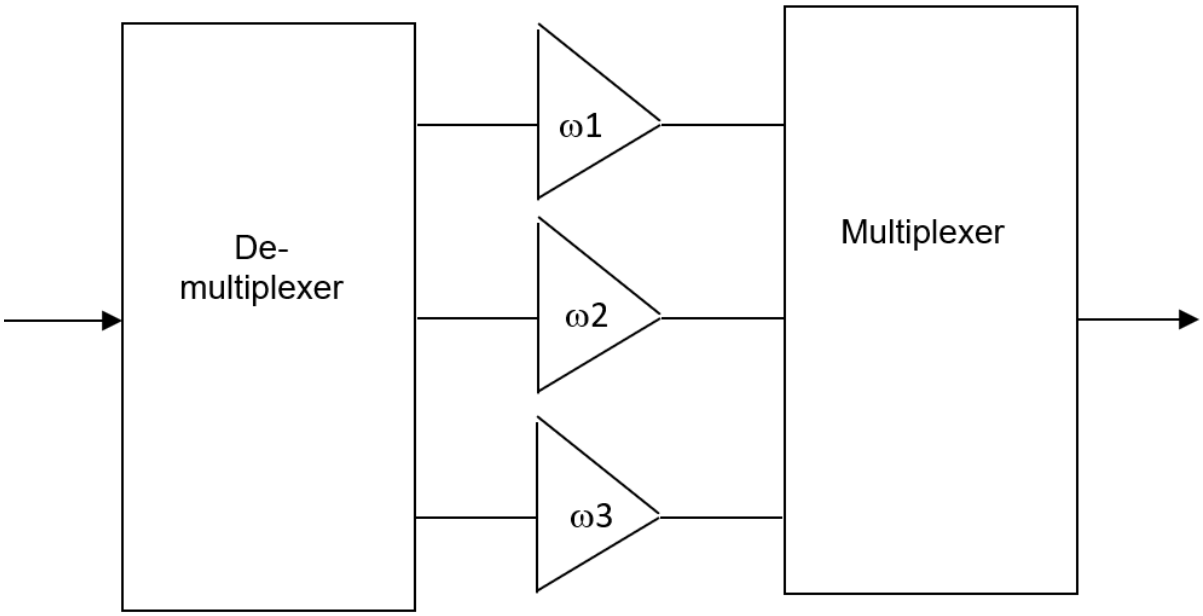


[7]

Model answer to Q 2(c): New Application

Employing the topologies given in 2(b), draw a unilateral 3-channel amplifier. Assume that for each frequency band the associated filters and amplifiers have the same -1 dB band edge frequencies.

ANSWERS



[4]

Model answer to Q 2(d): Calculated Example

Employing the topology from in 2(c), simple power loss budget analysis can be performed to calculate the overall gain, given the insertion loss of each circulator is 0.1 dB, the mid-band insertion loss of the filters is 0 dB and the mid-band gain of the amplifiers is 10 dB. State any simplifying assumptions.

(i) Calculate the gain at mid-band, assuming that the transition frequency bandwidth is insignificant.

-0.1 – 0 +10 – 0 -0.1 -0.1 – 0 -0.1 -0.1 -0 -0.1 = -9.4 dB

Total Insertion Loss = +9.4 dB

Assuming perfectly matched circulators and no inter-stage reflections between the filters and the amplifier

[3]

(ii) Calculate the gain at the -1 dB frequency band edge, assuming that the ideal upper -1 dB band edge frequency of one channel is the same as the ideal lower -1 dB band edge frequency of the adjacent channel.

-0.1-1 +9 -1 -0.1 -0.1 – 6.868 -0.1 -0.1-0 -0.1 = -0.468 dB

Total Insertion Loss = +0.468 dB

Assuming perfectly matched circulators and no inter-stage reflections between the filters and the amplifier, but taking into account the -6.868 dB return loss at the output of the final filter seen by the circulator.

[3]

(iii) Using the results from 2(d)(i) and (ii), comment on the resulting frequency response over the entire bandwidth of the 3 channels.

The frequency response for one channel amplifier overlaps the other. However, in the region of the overlapping -1 dB band edge, rather than a large power gain there is a power loss. As a result, the roll-off for each channel will be much steeper and so the overall ripple across the entire bandwidth of the 3 channels will be severe.

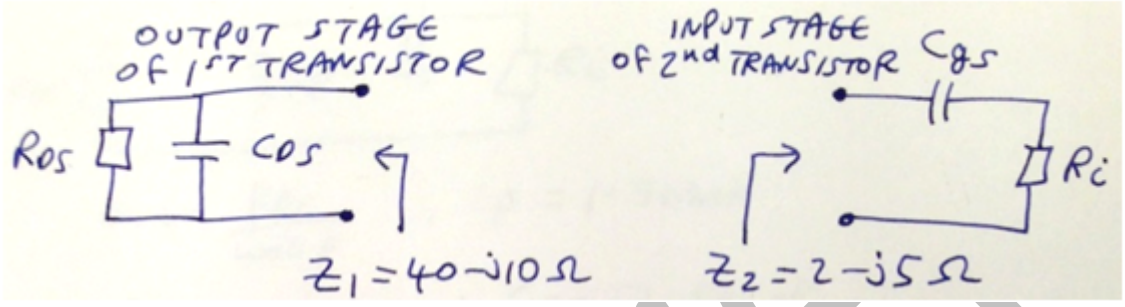
[2]

Answers

Model answer to Q 3(a): Calculated Example

A MESFET has input and output impedances of 2 – j 5 Ω and 40 – j 10 Ω, respectively. You are required to design the inter-stage impedance matching network for a two-stage 1 GHz amplifier. Assume that the MESFET input and output impedances are fixed and ignore DC biasing networks.

Draw the *RC* lumped-element models for the output of the first-stage transistor and the input for the second-stage transistor. Identify the dominant intrinsic parasitic resistive and capacitive elements of the MESFET.



[4]

Model answer to Q 3(b): Calculated Example

From the *RC* lumped-element models drawn in 3(a), calculate:

i) Parasitic values for both resistances and both capacitances.

$$Y_1 = \frac{1}{Z_1} = \frac{1}{R_1 + jX_1} \equiv G_1 + jB_1$$

$$\therefore R_{OS} = \frac{1}{G_1} = \frac{R_1^2 + X_1^2}{R_1} = 42.5\,\Omega$$

$$\text{and } C_{OS} = \frac{B_1}{\omega_0} = \frac{-X_1}{R_1^2 + X_1^2} = 0.936\,\text{pF}$$

$$Z_2 = R_2 + jX_2$$

$$\therefore R_i = R_2 = 2\,\Omega$$

$$\text{and } C_{GS} = \frac{-1}{\omega_0 X_2} = 31.831\,\text{pF}$$

[5]

ii) Using the values in 3(b)(i), calculate the loaded quality factors for the input and output of the MESFET. State if it is the input or output that limits the bandwidth of operation.

$Q_1 = \omega_0 C_{DS} R_{DS} = 0.25$

$Q_2 = \frac{1}{\omega_0 C_{gs} R_i} = 2.5$

SINCE $Q = \frac{f_0}{BW}$ AND $Q_1 < Q_2$

$\therefore$ INPUT LIMITS THE BANDWIDTH TO 0.4 GHz

[4]

Model answer to Q 3(c): Calculated Example

Using discrete lumped-element matching, calculate the:

i) lumped-element component values and state their orientation needed to resonate out the intrinsic parasitic capacitances of the MESFET at the output of the first MESFET and input of the second MESFET.

PUT A { $\frac{1}{C_{DS}\omega_0^2} = 27.06\,nH$ INDUCTOR IN PARALLEL WITH C_{DS}

$\frac{1}{C_{gs}\omega_0^2} = 0.796\,nH$ INDUCTOR IN SERIES WITH C_{gs}

[2]

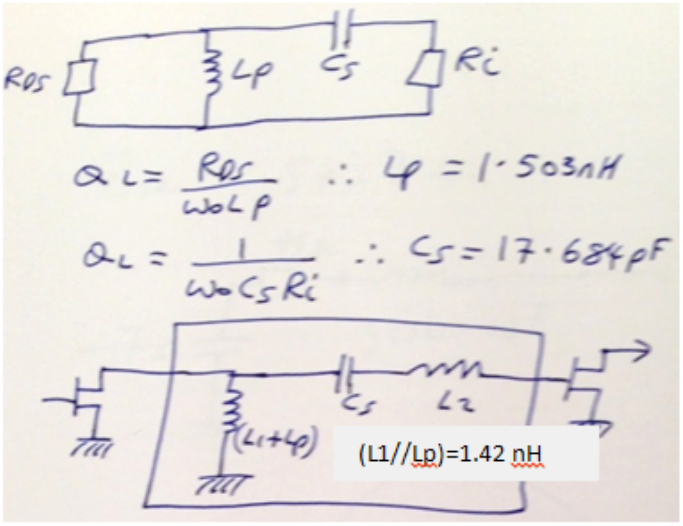
ii) loaded quality factor of the inter-stage impedance matching network and its bandwidth. State if it is the input of the first MESFET, inter-stage impedance matching network or output of the second MESFET that limits the bandwidth of operation.

$Q_L = \sqrt{\frac{R_{DS}}{R_i} - 1} = 4.5$

THIS NETWORK LIMITS THE BANDWIDTH TO 0.2 GHz

[2]

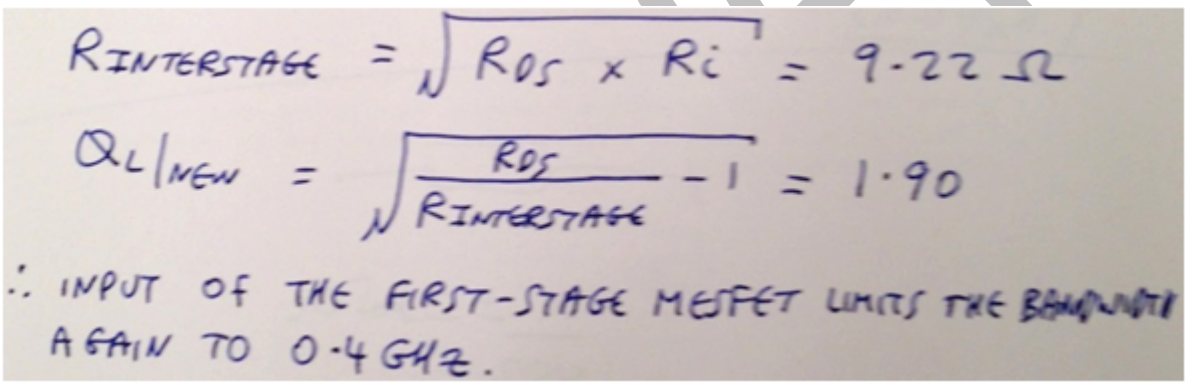
iii) lumped-element component values needed to implement a one-stage resistive L-matching network. Draw the complete impedance matching network having the minimal number of lumped-element components.



[4]

Model answer to Q 3(d): Calculated Example

With a two-stage resistive L-matching network calculate the new loaded quality factor. State if it is the input of the first MESFET, new inter-stage impedance matching network or output of the second MESFET that limits the bandwidth of operation.



[4]

Model answer to Q 4(a): Bookwork

Explain the difference between a duplexer and a diplexer.

A DUPLEXER is a 3-port network that allows the transceiver to use the same antenna.. A DIPLEXER is a 3-port filter that either separates or combines signals from 2 different frequency bands.

[2]

Model answer to Q 4(b): Bookwork

What is the most common way of implementing a duplexer in half-duplex and full-duplex mobile phone handsets?

In a half-duplex mobile phone handset, the duplexer is implemented with a transmit/receive (T/R) module, which normally employs a T/R switch (e.g. SP2T). In a full-duplex mobile phone handset, the duplexer can be as simple as a circulator in low-power applications or filters in high isolation applications (e.g. W-CDMA).

[4]

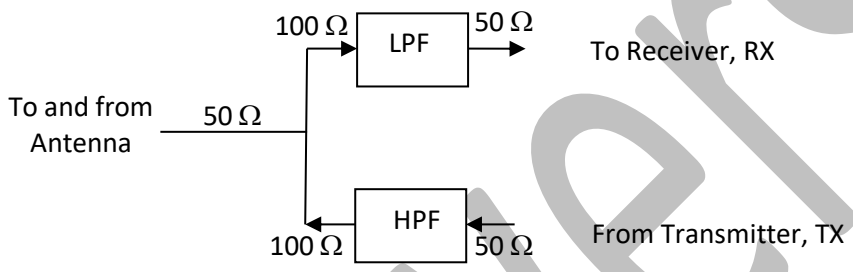
Model answer to Q 4(c): Calculated Application of Theory

Using the ideal lumped-element *LC* filter synthesis technique to implement a high pass filter and it associated low-pass prototype, design a simple 50 Ω GSM duplexer that can allow the following bands to pass through the duplexer with a 1 dB worst-case pass band attenuation ripple:

TX band: 930 ~ 943 MHz and RX band: 885 ~ 889 MHz

Draw the block diagram for this basic duplexer, indicating the directions of signal flow and any relevant impedances.

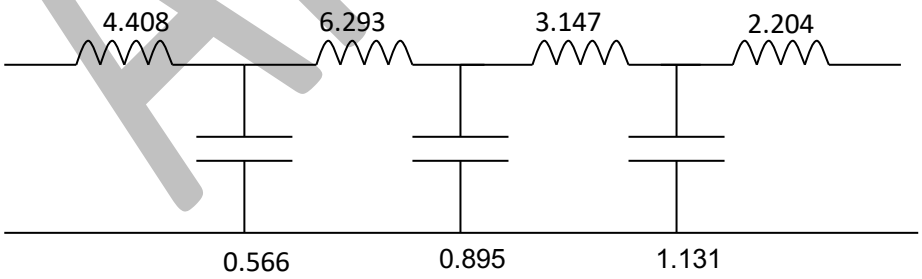
Hint: Choose 7th-order filters with a common cut-off frequency and common frequency ratio, giving approximately 10 dB rejection at the nearest band edge of the other band, and ensure that the one port impedance is double that of the other (e.g. the source termination impedance is Zs = 2Zo and the load termination impedance is Z_L = Zo = 50 Ω).



First choose a common cut-off frequency of the geometric mean between the upper end of the RX band and the lower cut-off frequency of the TX band (i.e. fc = sqrt(889 MHz x 930 MHz) = 909.27 MHz. Also, the frequency ratio for both filters is f/fc = 930/909.27 = 909.27/889 =1.023.

From graphs provided, for the Chebyshev low-pass filter with a 1 dB ripple, a 7th order filter is required to achieve an out-of-band rejection of ~ 10 dB and fc/f ratio of 1.023.

From tables provided, with R_S =2 R_L = 2Zo = 100, the prototype low-pass filter and associated coefficients are given below for a Chebyshev approximation:

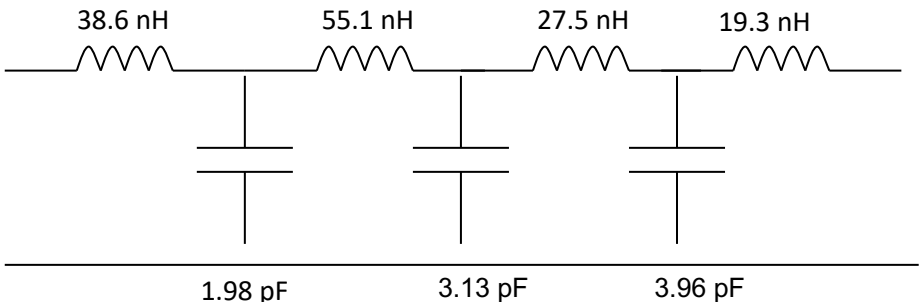


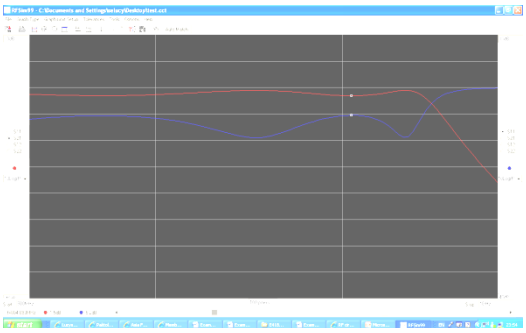
Low-pass de-normalising:

Series inductor: $L_S = \frac{R_L}{2\pi f_c} Ln$

Shunt Capacitor: $C_P = \frac{Cn}{2\pi f_c R_L}$

Answers





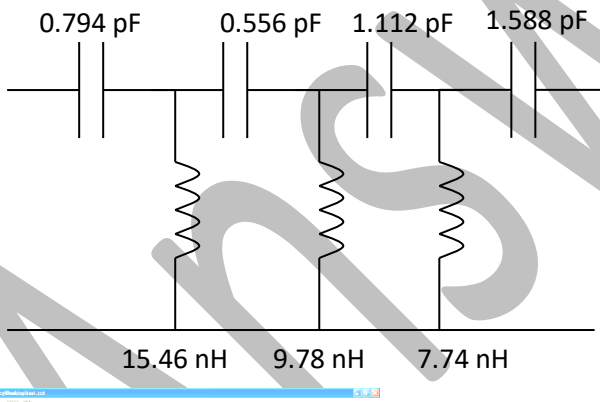
High-pass de-normalising:

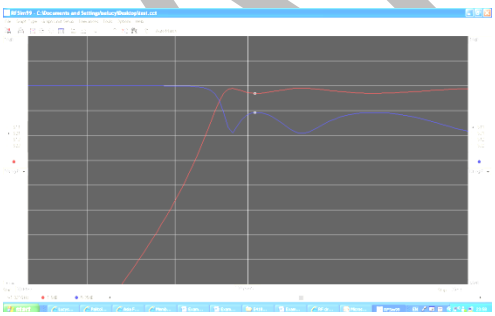
Shunt inductor:

$$L_P = \frac{R_L}{2\pi f_c L n}$$

Series Capacitor:

$$C_S = \frac{1}{2\pi f_c C n R_L}$$





The deviation in the results from those predicted by theory are due to limited component tolerances used within the simulations, giving rise to a 1.5 dB ripple instead of the desired 1 dB.

[12]

Model answer to Q 4(d): New Interpretation of Theory

Give four reasons why the design in 4(c) is not suitable for a real GSM mobile phone handset duplexer.

Answers

First of all, 7th order Chebyshev approximations for this very small frequency ratio only gives about 10 out-of-band rejection at the other band's nearest band edge frequency. This should be about 100 dB of rejection in practical mobile phone handset applications.

Also, with the transmitter, a BPF should be used instead of the HPF, to stop any harmonics from radiating out of the antenna.

Moreover, with the receiver, a BPF should be used instead of the LPF in order to minimise the noise bandwidth and, therefore, the noise power (as well as any interfering signals) entering into the receiver.

Finally, since the band-pass ripples are up 1 dB, the reflected power at each filter will effect the overall performance of the duplexer in both the TX and RX bands.

[4]

Model answer to Q 4(e): Calculated Application of Theory

Calculate the worst-case levels of return loss ripple in the pass bands for the ideal filters used in 4(c).

From the principle of conservation of energy, the worst-case return loss for a lossless LC filter will be $|S_{11}|^2 = 1-|S_{21}|^2 = -6.868$ dB, since the $|S_{21}|^2 = 0.79432$ for a 1 dB ripple.

[3]

Answers